

Worksheet 4 - Solution

PART 1: Laplace Transform Review

1) Obtain the Laplace transform of the following functions.

a) $x(t) = 15 + 3t^2$

$$X(s) = \frac{15}{s} + \frac{6}{s^3}$$

b) $x(t) = 8te^{-4t} + 2e^{-5t}$

$$X(s) = \frac{8}{(s+4)^2} + \frac{2}{s+5}$$

c) $x(t) = te^{-2t}\sin 4t$

If we define the $x(t) = ty(t)$, where $y(t) = e^{-2t}\sin 4t$. Then $X(s)$ is obtained as:

$$X(s) = -\frac{dY(s)}{ds}$$

Therefore, if $y(t) = e^{-2t}\sin 4t$, the $Y(s)$ is:

$$Y(s) = \frac{5}{(s+3)^2 + 5^2} = \frac{5}{s^2 + 6s + 34}$$

Then we can find the $X(s)$

$$X(s) = -\frac{dY(s)}{ds} = \frac{10s + 30}{(s^2 + 6s + 34)^2}$$

2) Use the initial-value and final-value theorems to determine $x(0)$ and $x(\infty)$ for the following transforms.

a) $X(s) = \frac{8}{2s+3}$

$$x(0) = \lim_{s \rightarrow \infty} sX(s) = \lim_{s \rightarrow \infty} s \left(\frac{8}{2s+3} \right) = \frac{8}{2}$$

$$x(\infty) = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} s \left(\frac{8}{2s+3} \right) = 0$$

b) $X(s) = \frac{7}{2s^2+6s+3}$

$$x(0) = \lim_{s \rightarrow \infty} sX(s) = \lim_{s \rightarrow \infty} s \left(\frac{7}{2s^2+6s+3} \right) = 0$$

$$x(\infty) = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} s \left(\frac{7}{2s^2+6s+3} \right) = 0$$

3) Obtain the inverse Laplace transform $x(t)$ for each of the following transforms.

a) $X(s) = \frac{3}{s(s+2)}$

$$X(s) = \frac{C_1}{s} + \frac{C_2}{s+2}$$

$$C_1 = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} \frac{3}{s+2} = \frac{3}{2}$$

$$C_2 = \lim_{s \rightarrow -2} (s+2)X(s) = \lim_{s \rightarrow -2} \frac{3}{s} = -\frac{3}{2}$$

$$X(s) = \frac{3/2}{s} + \frac{-3/2}{s+2} \rightarrow x(t) = \frac{3}{2}(1 - e^{-2t})$$

b) $X(s) = \frac{10s+7}{s(s+3)}$

$$X(s) = \frac{C_1}{s} + \frac{C_2}{s+3}$$

$$C_1 = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} \frac{10s+7}{s+3} = \frac{7}{3}$$

$$C_2 = \lim_{s \rightarrow -3} (s+3)X(s) = \lim_{s \rightarrow -3} \frac{10s+7}{s} = \frac{23}{3}$$

$$X(s) = \frac{7/3}{s} + \frac{23/3}{s+3} \rightarrow x(t) = \frac{7}{3} + \frac{23}{3}e^{-3t}$$

c) $X(s) = \frac{4s+7}{(s+2)(s+5)}$

$$X(s) = \frac{C_1}{s+2} + \frac{C_2}{s+5}$$

$$C_1 = \lim_{s \rightarrow -2} (s+2)X(s) = \lim_{s \rightarrow -2} \frac{4s+7}{s+5} = \frac{-1}{3}$$

$$C_2 = \lim_{s \rightarrow -5} (s+5)X(s) = \lim_{s \rightarrow -5} \frac{4s+7}{s+2} = \frac{13}{3}$$

$$X(s) = \frac{-1/3}{s+2} + \frac{13/3}{s+5} \rightarrow x(t) = -\frac{1}{3}e^{-2t} + \frac{13}{3}e^{-5t}$$

d) $X(s) = \frac{5}{s^2(s+4)}$

$$X(s) = \frac{C_1}{s} + \frac{C_2}{s^2} + \frac{C_3}{s+4}$$

$$C_1 = \lim_{s \rightarrow 0} \frac{d}{ds} [s^2 X(s)] = \lim_{s \rightarrow 0} \frac{d}{ds} \left[\frac{5}{s+4} \right] = \lim_{s \rightarrow 0} \frac{-5}{(s+4)^2} = \frac{-5}{16}$$

$$C_2 = \lim_{s \rightarrow 0} s^2 X(s) = \lim_{s \rightarrow 0} \frac{5}{s+4} = \frac{5}{4}$$

$$C_3 = \lim_{s \rightarrow -4} (s+4)X(s) = \lim_{s \rightarrow -4} \frac{5}{s^2} = \frac{5}{16}$$

$$X(s) = \frac{-5/16}{s} + \frac{5/4}{s^2} + \frac{5/16}{s+4} \rightarrow x(t) = -\frac{5}{16} + \frac{5}{4}t + \frac{5}{16}e^{-4t}$$

e) $X(s) = \frac{2}{s^2+16}$

$$X(s) = \frac{2}{s^2+4^2} = \frac{1}{2} \left(\frac{4}{s^2+4^2} \right) \rightarrow x(t) = \frac{1}{2} \sin 4t$$

f) $X(s) = \frac{7}{s^2+6s+13}$

$$X(s) = \frac{7}{(s^2+6s+9)+4} = \frac{7}{(s+3)^2+2^2} = \frac{7}{2} \left(\frac{2}{(s+3)^2+2^2} \right) \rightarrow x(t) = \frac{7}{2} e^{-3t} \sin 2t$$

g) $X(s) = \frac{2}{s(s^2+4s+13)}$

$$X(s) = \frac{C_1}{s} + \frac{C_2s+C_3}{s^2+4s+13}$$

$$C_1 = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} \frac{2}{s^2+4s+13} = \frac{2}{13}$$

$$2 = C_1(s^2+4s+13) + s(C_2s+C_3) \rightarrow 2 = (C_1+C_2)s^2 + (4C_1+C_3)s + 13C_1$$

$$C_1 = \frac{2}{13}, \quad C_2 = -\frac{2}{13}, \quad C_3 = -\frac{8}{13}$$

$$X(s) = \frac{\frac{2}{13}}{s} + \frac{-\frac{2}{13}s - \frac{8}{13}}{s^2+4s+13} \rightarrow X(s) = \frac{\frac{2}{13}}{s} + \frac{-\frac{2}{13}s - \frac{8}{13}}{(s^2+4s+4)+9} = \frac{\frac{2}{13}}{s} - \frac{2}{13} \left(\frac{s+4}{(s+2)^2+9} \right)$$

$$X(s) = \frac{\frac{2}{13}}{s} - \frac{2}{13} \left(\frac{s+2}{(s+2)^2+9} + \frac{2}{(s+2)^2+9} \right) = \frac{\frac{2}{13}}{s} - \frac{2}{13} \left(\frac{s+2}{(s+2)^2+9} \right) - \frac{4}{39} \left(\frac{3}{(s+2)^2+9} \right)$$

$$\rightarrow x(t) = \frac{2}{13} - \frac{2}{13} e^{-2t} \cos 3t - \frac{4}{39} e^{-2t} \sin 3t$$

h) $X(s) = \frac{16s^2+129s+200}{s(s+5)(s+8)}$

$$X(s) = \frac{C_1}{s} + \frac{C_2}{s+5} + \frac{C_3}{s+8}$$

$$C_1 = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} \frac{16s^2+129s+200}{(s+5)(s+8)} = \frac{200}{40} = 5$$

$$C_2 = \lim_{s \rightarrow -5} (s+5)X(s) = \lim_{s \rightarrow -5} \frac{16s^2+129s+200}{s(s+8)} = \frac{-45}{-15} = 3$$

$$C_3 = \lim_{s \rightarrow -8} (s+8)X(s) = \lim_{s \rightarrow -8} \frac{16s^2+129s+200}{s(s+5)} = \frac{192}{24} = 8$$

$$X(s) = \frac{5}{s} + \frac{3}{s+5} + \frac{8}{s+8} \rightarrow x(t) = 5 + 3e^{-5t} + 8e^{-8t}$$

$$i) \quad X(s) = \frac{12s^2 + 125s + 1268}{(s+7)(s^2 + 8s + 116)}$$

$$X(s) = \frac{C_1}{s+7} + \frac{C_2s + C_3}{s^2 + 8s + 116}$$

$$C_1 = \lim_{s \rightarrow -7} (s+7)X(s) = \lim_{s \rightarrow -7} \frac{12s^2 + 125s + 1268}{(s^2 + 8s + 116)} = \frac{981}{109} = 9$$

$$12s^2 + 125s + 1268 = C_1(s^2 + 8s + 116) + (s+7)(C_2s + C_3)$$

$$12s^2 + 125s + 1268 = (C_1 + C_2)s^2 + (8C_1 + 7C_2 + C_3)s + 116C_1 + 7C_3$$

$$C_1 = 9, \quad C_2 = 3, \quad C_3 = 32$$

$$X(s) = \frac{9}{s+7} + \frac{3s+32}{s^2+8s+116} \rightarrow X(s) = \frac{9}{s+7} + \frac{3s+32}{(s^2+8s+16)+100} = \frac{9}{s+7} + \frac{3s+12+20}{(s+4)^2+100}$$

$$X(s) = \frac{9}{s+7} + \frac{3(s+4)}{(s+4)^2+100} + \frac{2(10)}{(s+4)^2+100}$$

$$\rightarrow x(t) = 9e^{-7t} + 3e^{-4t} \cos 10t + 2e^{-4t} \sin 10t$$

4) Use the Laplace transform to solve the following ordinary differential equations.

a) $5\dot{x}(t) + 7x(t) = 0, \quad x(0) = 4$

$$5(sX(s) - x(0)) + 7X(s) = 0 \rightarrow 5(sX(s) - 4) + 7X(s) = 0 \rightarrow (5s+7)X(s) - 20 = 0$$

$$\rightarrow X(s) = \frac{20}{5s+7} \rightarrow x(t) = 4e^{-7t/5}$$

b) $5\dot{x}(t) + 7x(t) = 15, \quad x(0) = 0$

$$5(sX(s) - x(0)) + 7X(s) = \frac{15}{s} \rightarrow (5s+7)X(s) = \frac{15}{s}$$

$$\rightarrow X(s) = \frac{15}{s(5s+7)} = \frac{3}{s(s+7/5)} = \frac{15/7}{s} + \frac{-15/7}{s+7/5} \rightarrow x(t) = \frac{15}{7} - \frac{15}{7}e^{-7t/5}$$

c) $\dot{x}(t) + 7x(t) = 4t, \quad x(0) = 5$

$$sX(s) - x(0) + 7X(s) = \frac{4}{s^2} \rightarrow sX(s) - 5 + 7X(s) = \frac{4}{s^2} \rightarrow (s+7)X(s) = \frac{4}{s^2} + 5$$

$$\rightarrow X(s) = \frac{5s^2+4}{s^2(s+7)} \rightarrow X(s) = \frac{4/49}{s} + \frac{4/7}{s^2} + \frac{249/49}{s+7} \rightarrow x(t) = \frac{4}{49} + \frac{4}{7}t + \frac{249}{49}e^{-7t}$$

d) $\ddot{x}(t) + 7\dot{x}(t) + 10x(t) = 20, \quad x(0) = 5, \quad \dot{x}(0) = 3$

$$s^2X(s) - sx(0) - \dot{x}(0) + 7(sX(s) - x(0)) + 10X(s) = \frac{20}{s}$$

$$\rightarrow s^2X(s) - 5s - 3 + 7sX(s) - 35 + 10X(s) = \frac{20}{s} \rightarrow (s^2 + 7s + 10)X(s) = \frac{20}{s} + 5s + 38$$

$$\rightarrow X(s) = \frac{5s^2 + 38s + 20}{s(s^2 + 7s + 10)} = \frac{5s^2 + 38s + 20}{s(s+2)(s+5)} = \frac{2}{s} + \frac{6}{s+2} + \frac{-3}{s+5}$$

$$\rightarrow x(t) = 2 + 6e^{-2t} - 3e^{-5t}$$

e) $5\ddot{x}(t) + 20\dot{x}(t) + 20x(t) = 28, \quad x(0) = 5, \quad \dot{x}(0) = 8$

$$5(s^2X(s) - sx(0) - \dot{x}(0)) + 20(sX(s) - x(0)) + 20X(s) = \frac{28}{s}$$

$$\rightarrow 5s^2X(s) - 25s - 40 + 20sX(s) - 100 + 20X(s) = \frac{28}{s}$$

$$\rightarrow (5s^2 + 20s + 20)X(s) = \frac{28}{s} + 25s + 140$$

$$\rightarrow X(s) = \frac{25s^2 + 140s + 28}{s(5s^2 + 20s + 20)} = \frac{25s^2 + 140s + 28}{5s(s+2)^2} = \frac{7/5}{s} + \frac{18/5}{s+2} + \frac{76/5}{(s+2)^2}$$

$$\rightarrow x(t) = \frac{7}{5} + \frac{18}{5}e^{-2t} + \frac{76}{5}te^{-2t}$$

f) $\ddot{x}(t) + 16x(t) = 144, \quad x(0) = 5, \quad \dot{x}(0) = 12$

$$s^2X(s) - sx(0) - \dot{x}(0) + 16X(s) = \frac{144}{s} \rightarrow s^2X(s) - 5s - 12 + 16X(s) = \frac{144}{s}$$

$$\rightarrow (s^2 + 16)X(s) = \frac{144}{s} + 5s + 12$$

$$\rightarrow X(s) = \frac{5s^2 + 12s + 144}{s(s^2 + 16)} = \frac{9}{s} + \frac{-4s + 12}{s^2 + 16} = \frac{9}{s} + \frac{-4s}{s^2 + 16} + \frac{12}{s^2 + 16}$$

$$\rightarrow x(t) = \frac{7}{5} - 4 \cos 4t + 3 \sin 4t$$

g) $\ddot{x}(t) + 14\dot{x}(t) + 49x(t) = 0, \quad x(0) = 1, \quad \dot{x}(0) = 3$

$$s^2X(s) - sx(0) - \dot{x}(0) + 14(sX(s) - x(0)) + 49X(s) = 0$$

$$\rightarrow s^2X(s) - s - 3 + 14sX(s) - 14 + 49X(s) = 0 \rightarrow (s^2 + 14s + 49)X(s) = s + 17$$

$$\rightarrow X(s) = \frac{s + 17}{s^2 + 14s + 49} = \frac{s + 17}{(s + 7)^2} = \frac{1}{s + 7} + \frac{10}{(s + 7)^2}$$

$$\rightarrow x(t) = e^{-7t} + 10te^{-7t}$$

PART 2: Transfer Function from Physical System

1) In the following mass-spring-damper systems, the input is the displacement y and the output is the displacements x of the mass m . The equilibrium positions corresponds to $x = y = 0$. Neglect the friction between the mass and the surface. Derive the equations of motion and find the transfer function $X(s)/Y(s)$.

Direction of motion is to the right.

The equation of motion is obtained as:

$$\text{Mass } m \rightarrow -k_2x - k_1(x - y) - c(\dot{x} - \dot{y}) = m\ddot{x}$$

Rearrange the equation of motion:

$$m\ddot{x} + c\dot{x} + (k_1 + k_2)x = c\dot{y} + k_1y$$

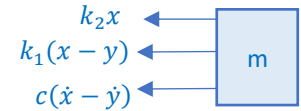
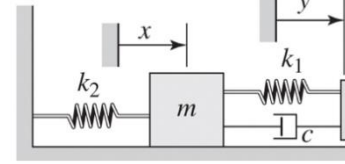
Take Laplace transform by considering the zero initial conditions:

$$ms^2X(s) + csX(s) + (k_1 + k_2)X(s) = csY(s) + k_1Y(s)$$

$$(ms^2 + cs + (k_1 + k_2))X(s) = (cs + k_1)Y(s)$$

The transfer function is:

$$\frac{X(s)}{Y(s)} = \frac{cs + k_1}{ms^2 + cs + k_1 + k_2}$$



2) Find the transfer function $Z(s)/X(s)$ for the system shown below.

Direction of motion is downward.

The equation of motions for massless points A and B are obtained as:

$$\text{Point A} \rightarrow -k_1(y - x) - c(\dot{y} - \dot{z}) = 0 \rightarrow c\dot{y} + k_1y - c\dot{z} - k_1x = 0$$

$$\text{Point B} \rightarrow -c(\dot{z} - \dot{y}) - k_2z = 0 \rightarrow c\dot{y} - c\dot{z} - k_2z = 0$$

Take Laplace transform by considering the zero initial conditions:

$$csY(s) + k_1Y(s) - csZ(s) - k_1X(s) = 0 \rightarrow (cs + k_1)Y(s) - csZ(s) - k_1X(s) = 0 \quad \text{Eqn. (1)}$$

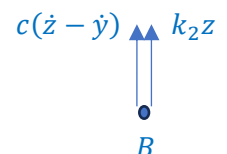
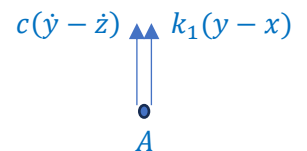
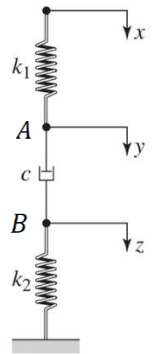
$$csY(s) - csZ(s) - k_2Z(s) = 0 \rightarrow Y(s) = \frac{cs + k_2}{cs}Z(s) \quad \text{Eqn. (2)}$$

Substitute $Y(s)$ from Eqn. (2) into the Eqn. (1) and find the transfer function $Z(s)/X(s)$:

$$(cs + k_1)\frac{cs + k_2}{cs}Z(s) - csZ(s) - k_1X(s) = 0$$

$$\left((cs + k_1)\frac{cs + k_2}{cs} - cs \right) Z(s) - k_1X(s) = 0$$

$$\rightarrow \frac{Z(s)}{X(s)} = \frac{ck_1s}{(k_1 + k_2)cs + k_1k_2}$$



9) Find the transfer function $Y(s)/X(s)$ for the system shown below.

Direction of motion is downward.

The equation of motion is obtained as:

$$\text{Point } A \rightarrow -k(y-x) - c_1(\dot{y}-\dot{x}) - c_2\dot{y} = 0$$

$$(c_1 + c_2)\dot{y} + ky - c_1\dot{x} - kx = 0$$

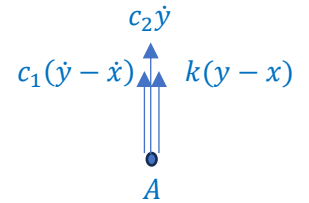
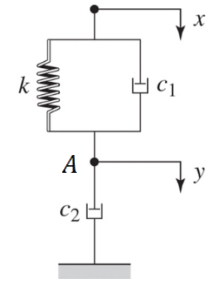
Take Laplace transform by considering the zero initial conditions:

$$(c_1 + c_2)sY(s) + kY(s) - c_1sX(s) - kX(s) = 0$$

$$((c_1 + c_2)s + k)Y(s) = (c_1s + k)X(s)$$

Find the transfer function $Y(s)/X(s)$:

$$\frac{Y(s)}{X(s)} = \frac{c_1s + k}{(c_1 + c_2)s + k}$$



10) Find the transfer function $Y(s)/X(s)$ for the system shown below.

Direction of motion is downward.

The equation of motions are obtained as:

$$\text{Point } A \rightarrow -k_1(y-x) - c_1(\dot{y}-\dot{x}) - c_2(\dot{y}-\dot{z}) = 0$$

$$(c_1 + c_2)\dot{y} + k_1y - c_1\dot{x} - k_1x - c_2\dot{z} = 0$$

$$\text{Point } B \rightarrow -c_2(\dot{z}-\dot{y}) - k_2z = 0 \rightarrow c_2\dot{z} + k_2z - c_2\dot{y} = 0$$

Take Laplace transform by considering the zero initial conditions:

$$(c_1 + c_2)sY(s) + k_1Y(s) - c_1sX(s) - k_1X(s) - c_2sZ(s) = 0 \quad \text{Eqn. (1)}$$

$$c_2sZ(s) + k_2Z(s) - c_2sY(s) = 0 \rightarrow Z(s) = \frac{c_2s}{c_2s + k_2} Y(s) \quad \text{Eqn. (2)}$$

Rearrange the Eqn. (1) and substitute $Z(s)$ from Eqn. (2) into the Eqn. (1):

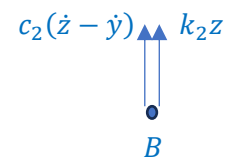
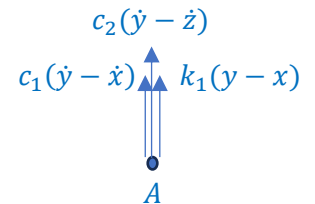
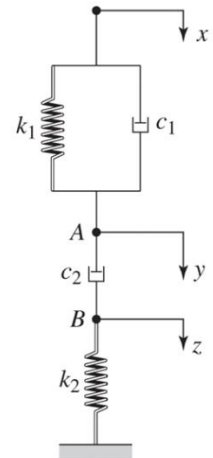
$$((c_1 + c_2)s + k_1)Y(s) - (c_1s + k_1)X(s) - c_2sZ(s) = 0$$

$$((c_1 + c_2)s + k_1)Y(s) - (c_1s + k_1)X(s) - c_2s \frac{c_2s}{c_2s + k_2} Y(s) = 0$$

$$\left((c_1 + c_2)s + k_1 - c_2s \frac{c_2s}{c_2s + k_2} \right) Y(s) = (c_1s + k_1)X(s)$$

Find the transfer function $Y(s)/X(s)$:

$$\frac{Y(s)}{X(s)} = \frac{(c_1s + k_1)(c_2s + k_2)}{c_1c_2s^2 + ((c_1 + c_2)k_2 + k_1c_2)s + k_1k_2}$$



11) In the system shown below, the input is the force f and the output is the displacement x_A of point A. When $x = x_A$, the spring is at its free length. Derive the equation of motion and transfer function $X_A(s)/F(s)$.

Direction of motion is to the right.

The equation of motion is obtained as:

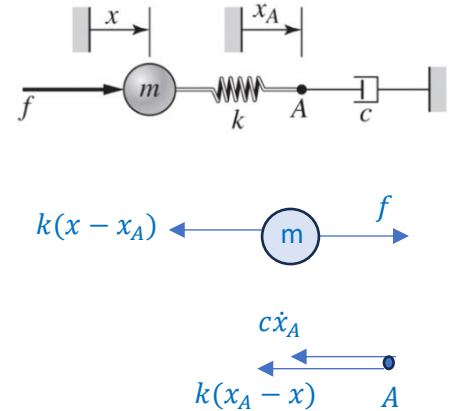
$$\text{Mass } m \rightarrow f(t) - k(x - x_A) = m\ddot{x} \rightarrow f(t) = m\ddot{x} + kx - kx_A$$

$$\text{Point A} \rightarrow -k(x_A - x) - c\dot{x}_A = 0 \rightarrow c\dot{x}_A + kx_A = kx$$

Take Laplace transform by considering the zero initial conditions:

$$F(s) = ms^2X(s) + kX(s) - kX_A(s) \quad \text{Eqn. (1)}$$

$$csX_A(s) + kX_A(s) = kX(s) \rightarrow X(s) = \frac{cs + k}{k}X_A(s) \quad \text{Eqn. (2)}$$



Substitute $X(s)$ from Eqn. (2) into the Eqn. (1), and form the transfer function:

$$F(s) = (ms^2 + k)\frac{cs + k}{k}X_A(s) - kX_A(s) \rightarrow F(s) = \left((k + ms^2)\frac{cs + k}{k} - k \right)X_A(s)$$

$$\frac{X_A(s)}{F(s)} = \frac{k}{s(mcs^2 + mks + ck)}$$

12) In the system shown below, the input is the displacement y and the output is the displacement x . When $x = y = 0$, the springs are at their free length. Derive the equation of motion and transfer function $X(s)/Y(s)$.

Direction of motion is to the right.

Displacement of the massless point A is x_A .

The equation of motion is obtained as:

$$\text{Mass } m \rightarrow -c(\dot{x} - \dot{x}_A) - k_2x = m\ddot{x} \rightarrow m\ddot{x} + c\dot{x} + k_2x - c\dot{x}_A = 0$$

$$\text{Point A} \rightarrow -k_1(x_A - y) - c(\dot{x}_A - \dot{x}) = 0 \rightarrow c\dot{x}_A + k_1x_A = c\dot{x} + k_1y$$

Take Laplace transform by considering the zero initial conditions:

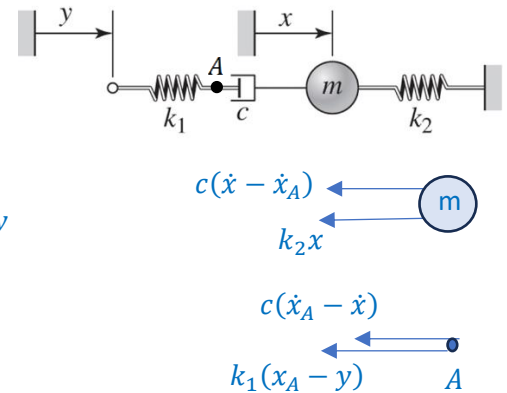
$$ms^2X(s) + csX(s) + k_2X(s) - csX_A(s) = 0 \quad \text{Eqn. (1)}$$

$$csX_A(s) + k_1X_A(s) = csX(s) + k_1Y(s) \rightarrow X_A(s) = \frac{cs}{cs + k_1}X(s) + \frac{k_1}{cs + k_1}Y(s) \quad \text{Eqn. (2)}$$

Substitute $X_A(s)$ from Eqn. (2) into the Eqn. (1), and form the transfer function:

$$(ms^2 + cs + k_2)X(s) - csX_A(s) = 0 \rightarrow (ms^2 + cs + k_2)X(s) - \frac{cs^2}{cs + k_1}Y(s) - \frac{c^2s^2}{cs + k_1}X(s) = 0$$

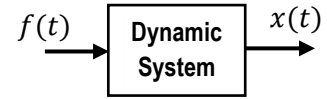
$$\frac{X(s)}{Y(s)} = \frac{ck_1s}{mcs^3 + mk_1s^2 + c(k_1 + k_2)s + k_1k_2}$$



PART 3: Transfer Functions from Differential Equations

1) Assume the following differential equation model of SISO systems with $f(t)$ as the input and $x(t)$ as the output. Determine the transfer function $X(s)/F(s)$ and compute the poles.

a) $10\dot{x}(t) + 14x(t) = 15f(t)$



Take Laplace transform of the equation, assuming zero initial condition

$$10sX(s) + 14X(s) = 15F(s) \rightarrow \frac{X(s)}{F(s)} = \frac{15}{10s + 14}$$

Poles are roots of the characteristic equation, which is the denominator polynomial.

$$10s + 14 = 0 \rightarrow s = -\frac{14}{10} = -1.4$$

Poles: $s = -1.4$

b) $6\ddot{x}(t) + 60\dot{x}(t) + 126x(t) = 7f(t)$

Take Laplace transform of the equation, assuming zero initial condition

$$6s^2X(s) + 60sX(s) + 126X(s) = 7F(s) \rightarrow \frac{X(s)}{F(s)} = \frac{7}{6s^2 + 60s + 126}$$

Poles are roots of the characteristic equation, which is the denominator polynomial.

$$6s^2 + 60s + 126 = 0 \rightarrow s^2 + 10s + 21 = 0 \rightarrow (s + 7)(s + 3) = 0$$

Poles: $s_1 = -7, s_2 = -3$

c) $4\ddot{x}(t) + 56\dot{x}(t) + 232x(t) = 8\dot{f}(t) + 3f(t)$

Take Laplace transform of the equation, assuming zero initial condition

$$4s^2X(s) + 56sX(s) + 232X(s) = 8sF(s) + 3F(s) \rightarrow \frac{X(s)}{F(s)} = \frac{8s + 3}{4s^2 + 56s + 232}$$

Poles are roots of the characteristic equation, which is the denominator polynomial.

$$4s^2 + 56s + 232 = 0 \rightarrow s^2 + 14s + 58 = 0$$

Poles: $s_{1,2} = -7 \pm 3j$

d) $10\dot{x}(t) + 14x(t) = 6\dot{f}(t) + 15f(t)$

Take Laplace transform of the equation, assuming zero initial condition

$$10sX(s) + 14X(s) = 6sF(s) + 15F(s) \rightarrow \frac{X(s)}{F(s)} = \frac{6s + 15}{10s + 14}$$

Poles are roots of the characteristic equation, which is the denominator polynomial.

$$10s + 14 = 0 \rightarrow s = -\frac{14}{10} = -1.4$$

Poles: $s = -1.4$

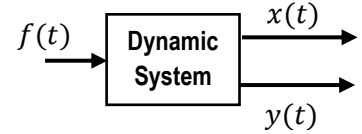
2) Assume the following differential equation model of MIMO systems with $f(t)$ as the input and $x(t)$ and $y(t)$ as the outputs. Obtain the transfer functions $X(s)/F(s)$ and $Y(s)/F(s)$ for each of the following models.

a) $4\dot{x}(t) = y(t), \quad \dot{y}(t) = f(t) - 5y(t) - 17x(t)$

Take Laplace transform of each equation, assuming zero initial condition

$$4sX(s) = Y(s) \quad \text{Eqn. (1)}$$

$$sY(s) = F(s) - 5Y(s) - 17X(s) \rightarrow (s + 5)Y(s) = F(s) - 17X(s) \quad \text{Eqn. (2)}$$



Substitute $Y(s)$ from Eqn. (1) into Eqn. (2):

$$4s(s + 5)X(s) = F(s) - 17X(s) \rightarrow (4s^2 + 20s + 17)X(s) = F(s)$$

$$\frac{X(s)}{F(s)} = \frac{1}{4s^2 + 20s + 17}$$

From Eqn. (1):

$$X(s) = \frac{1}{4s} Y(s) \quad \text{Eqn. (3)}$$

Substitute $X(s)$ from Eqn. (3) into Eqn. (2):

$$(s + 5)Y(s) = F(s) - \frac{17}{4s} Y(s) \rightarrow 4s(s + 5)Y(s) = 4sF(s) - 17Y(s)$$

$$(4s^2 + 20s + 17)X(s) = 4sF(s) \rightarrow \frac{Y(s)}{F(s)} = \frac{4s}{4s^2 + 20s + 17}$$

b) $\dot{x}(t) = -3x(t) + 7y(t), \quad \dot{y}(t) = f(t) - 8y(t) - 3x(t)$

Take Laplace transform of each equation, assuming zero initial condition

$$sX(s) = -3X(s) + 7Y(s) \rightarrow (s + 3)X(s) = 7Y(s) \quad \text{Eqn. (1)}$$

$$sY(s) = F(s) - 8Y(s) - 3X(s) \rightarrow (s + 8)Y(s) = F(s) - 3X(s) \quad \text{Eqn. (2)}$$

From Eqn. (1) find $X(s)$:

$$X(s) = \frac{7}{s + 3} Y(s) \quad \text{Eqn. (3)}$$

Substitute $X(s)$ from Eqn. (3) into Eqn. (2):

$$(s + 8)Y(s) = F(s) - \frac{21Y(s)}{s + 3} \rightarrow (s + 8)(s + 3)Y(s) = (s + 3)F(s) - 21Y(s)$$

$$(s^2 + 3s + 8s + 24 + 21)Y(s) = (s + 3)F(s) \rightarrow \frac{Y(s)}{F(s)} = \frac{s + 3}{s^2 + 11s + 45}$$

From Eqn. (1) find $Y(s)$:

$$Y(s) = \frac{s+3}{7}X(s) \quad \text{Eqn. (4)}$$

Substitute $Y(s)$ from Eqn. (4) into Eqn. (2):

$$(s+8)\left(\frac{s+3}{7}\right)X(s) = F(s) - 3X(s) \rightarrow (s+8)(s+3)X(s) = 7F(s) - 21X(s)$$

$$(s^2 + 3s + 8s + 24 + 21)Y(s) = 7F(s) \rightarrow \frac{X(s)}{F(s)} = \frac{7}{s^2 + 11s + 45}$$

c) $4\dot{x}(t) = y(t), \quad \dot{y}(t) = f(t) - 3y(t) - 12x(t)$

Take Laplace transform of each equation, assuming zero initial condition

$$4sX(s) = Y(s) \quad \text{Eqn. (1)}$$

$$sY(s) = F(s) - 3Y(s) - 12X(s) \rightarrow (s+3)Y(s) = F(s) - 12X(s) \quad \text{Eqn. (2)}$$

Substitute $Y(s)$ from Eqn. (1) into Eqn. (2):

$$4s(s+3)X(s) = F(s) - 12X(s) \rightarrow (4s^2 + 12s + 12)X(s) = F(s)$$

$$\frac{X(s)}{F(s)} = \frac{1}{4s^2 + 12s + 12}$$

From Eqn. (1) find $X(s)$:

$$X(s) = \frac{Y(s)}{4s} \quad \text{Eqn. (3)}$$

Substitute $X(s)$ from Eqn. (3) into Eqn. (2):

$$(s+3)Y(s) = F(s) - \frac{12}{4s}Y(s) \rightarrow 4s(s+3)Y(s) = 4sF(s) - 12Y(s)$$

$$(4s^2 + 12s + 12)Y(s) = 4sF(s) \rightarrow \frac{Y(s)}{F(s)} = \frac{4s}{4s^2 + 12s + 12}$$

PART 4: State-Space Model from Transfer Function Model

1) Obtain the state-space model for the following transfer function models. Select the state variables in phase-variable form. Draw the equivalent block diagram showing the state variables, input and output of the system.

a) $\frac{Y(s)}{F(s)} = \frac{4}{9s^2 + 2s + 7}$

This is a **strictly proper** transfer function with a **constant value** in the numerator.

First, find the associate differential equation:

$$9s^2Y(s) + 2sY(s) + 7Y(s) = 4F(s) \quad \rightarrow \quad 9\ddot{y}(t) + 2\dot{y}(t) + 7y(t) = 4f(t)$$

Define the state variables:

$$q_1(t) = y(t)$$

$$q_2(t) = \dot{y}(t)$$

Find the first derivative of the state variables and rewrite them in terms of the state variables and the input.

$$\dot{q}_1(t) = \dot{y}(t) \quad \rightarrow \quad \dot{q}_1(t) = q_2(t)$$

$$\dot{q}_2(t) = \ddot{y}(t) \quad \rightarrow \quad \dot{q}_2(t) = \frac{1}{9}(4f(t) - 2\dot{y}(t) - 7y(t)) = \frac{4}{9}f(t) - \frac{2}{9}q_2(t) - \frac{7}{9}q_1(t)$$

Find the output in terms of the state variables and the input.

$$y(t) = q_1(t)$$

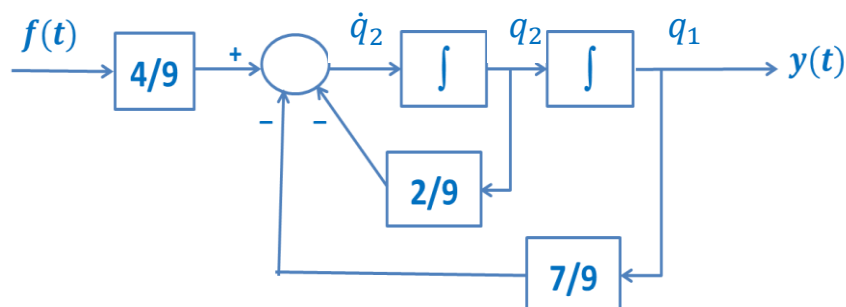
The system model has 2 state variables, 1 input, and 1 output.

Form the state equations and the output equations in the standard matrix-vector form.

$$\text{State Equation} \quad \rightarrow \quad \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{7}{9} & -\frac{2}{9} \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{4}{9} \end{bmatrix} f(t)$$

$$\text{Output Equation} \quad \rightarrow \quad y(t) = [1 \quad 0] \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + [0]f(t)$$

The block diagram to visualize the state variables, input, and output:



b) $\frac{Y(s)}{F(s)} = \frac{4s+7}{s+5}$

This is a **proper** transfer function.

First, we have to rewrite it as a summation of a **constant term** and a **strictly proper** function.

$$\frac{Y(s)}{F(s)} = \frac{4s+7}{s+5} = 4 + \frac{-13}{s+5}$$

The feed-forward matrix **D** is obtained as the constant term 4.

Then, determine the matrices **A**, **B**, and **C** from the strictly proper transfer function.

$$\frac{Z(s)}{F(s)} = \frac{-13}{s+5}$$

First, find the associate differential equation:

$$sZ(s) + 5Z(s) = -13F(s) \rightarrow \dot{z}(t) + 5z(t) = -13f(t)$$

Define the state variable:

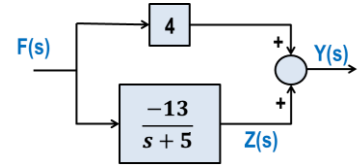
$$q_1(t) = z(t)$$

Find the first derivative of the state variable and rewrite it in terms of the state variable and the input.

$$\dot{q}_1(t) = \dot{z}(t) \rightarrow \dot{q}_1(t) = -5q_1(t) - 13f(t)$$

Find the output in terms of the state variables and the input.

$$y(t) = 4f(t) + z(t) = 4f(t) + q_1(t)$$



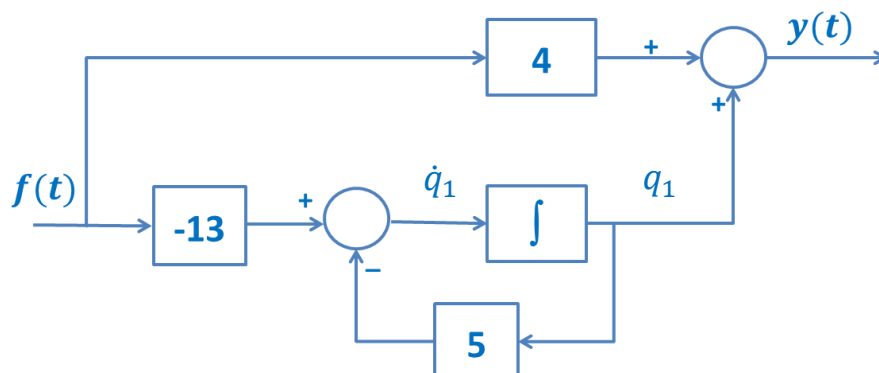
The system model has 1 state variable, 1 input, and 1 output.

Form the state equations and the output equations in the standard matrix-vector form.

State Equation $\rightarrow \dot{q}_1(t) = [-5]q_1(t) + [-13]f(t)$

Output Equation $\rightarrow y(t) = [1]q_1(t) + [4]f(t)$

The block diagram to visualize the state variable, input, and output:

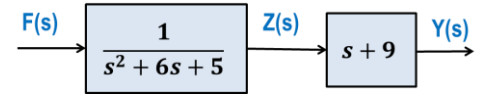


c) $\frac{Y(s)}{F(s)} = \frac{s+9}{s^2+6s+5}$

This is a **strictly proper** transfer function with a **polynomial** in the numerator.

Since, the numerator is a polynomial of s , we have to split it into two parts as below:

$$\frac{Y(s)}{F(s)} = \frac{s+9}{s^2+6s+5} = \left(\frac{1}{s^2+6s+5} \right) (s+9)$$



First, find the state equation from the part with the denominator:

$$\frac{Z(s)}{F(s)} = \frac{1}{s^2+6s+5}$$

$$s^2 Z(s) + 6s Z(s) + 5Z(s) = F(s) \quad \rightarrow \quad \ddot{z}(t) + 6\dot{z}(t) + 5z(t) = f(t)$$

Define the state variables:

$$q_1(t) = z(t)$$

$$q_2(t) = \dot{z}(t)$$

Find the first derivative of the state variables and rewrite them in terms of the state variables and the input.

$$\dot{q}_1(t) = \dot{z}(t) \quad \rightarrow \quad \dot{q}_1(t) = q_2(t)$$

$$\dot{q}_2(t) = \ddot{z}(t) \quad \rightarrow \quad \dot{q}_2(t) = -6\dot{z}(t) - 5z(t) + f(t) = -6q_2(t) - 5q_1(t) + f(t)$$

Find the output equation by considering the effect of the block with the numerator.

$$Y(s) = (s+9)Z(s) \quad \rightarrow \quad Y(s) = sZ(s) + 9Z(s) \quad \rightarrow \quad y(t) = \dot{z}(t) + 9z(t) = q_2(t) + 9q_1(t)$$

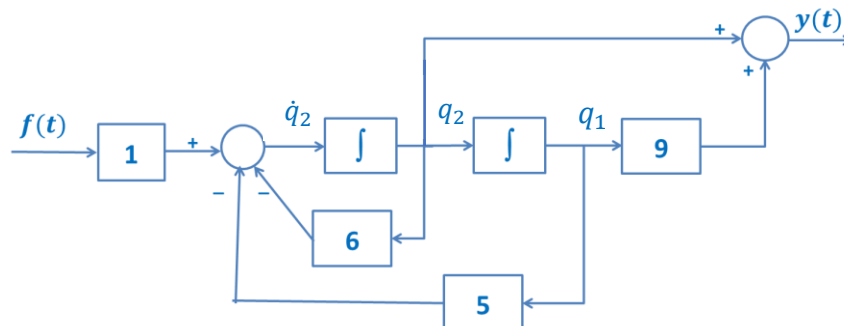
The system model has 2 state variables, 1 input, and 1 output.

Form the state equations and the output equations in the standard matrix-vector form.

$$\text{State Equation} \quad \rightarrow \quad \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -5 & -6 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f(t)$$

$$\text{Output Equation} \quad \rightarrow \quad y(t) = \begin{bmatrix} 9 & 1 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} f(t)$$

The block diagram to visualize the state variables, input, and output:

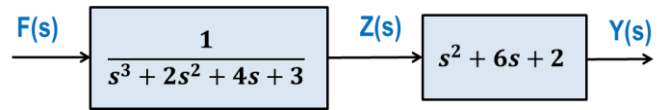


d) $\frac{Y(s)}{F(s)} = \frac{s^2 + 6s + 2}{s^3 + 2s^2 + 4s + 3}$

This is a **strictly proper** transfer function with a **polynomial** in the numerator.

Since, the numerator is a polynomial of s , we have to split it into two parts as below:

$$\frac{Y(s)}{F(s)} = \frac{s^2 + 6s + 2}{s^3 + 2s^2 + 4s + 3} = \left(\frac{1}{s^3 + 2s^2 + 4s + 3} \right) (s^2 + 6s + 2)$$



First, find the state equation from the part with the denominator:

$$\frac{Z(s)}{F(s)} = \frac{1}{s^3 + 2s^2 + 4s + 3}$$

$$s^3 Z(s) + 2s^2 Z(s) + 4s Z(s) + 3Z(s) = F(s) \quad \rightarrow \quad \ddot{z}(t) + 2\dot{z}(t) + 4z(t) + 3z(t) = f(t)$$

Define the state variables:

$$q_1(t) = z(t)$$

$$q_2(t) = \dot{z}(t)$$

$$q_3(t) = \ddot{z}(t)$$

Find the first derivative of the state variables and rewrite them in terms of the state variables and the input.

$$\dot{q}_1(t) = \dot{z}(t) \quad \rightarrow \quad \dot{q}_1(t) = q_2(t)$$

$$\dot{q}_2(t) = \ddot{z}(t) \quad \rightarrow \quad \dot{q}_2(t) = q_3(t)$$

$$\dot{q}_3(t) = \dddot{z}(t) \quad \rightarrow \quad \dot{q}_3(t) = -2\ddot{z}(t) - 4\dot{z}(t) - 3z(t) + f(t) = -2q_3(t) - 4q_2(t) - 3q_1(t) + f(t)$$

Find the output equation by considering the effect of the block with the numerator.

$$Y(s) = (s^2 + 6s + 2)Z(s) \quad \rightarrow \quad Y(s) = s^2 Z(s) + 6s Z(s) + 2Z(s)$$

$$\rightarrow y(t) = \ddot{z}(t) + 6\dot{z}(t) + 2z(t) = q_3(t) + 6q_2(t) + 2q_1(t)$$

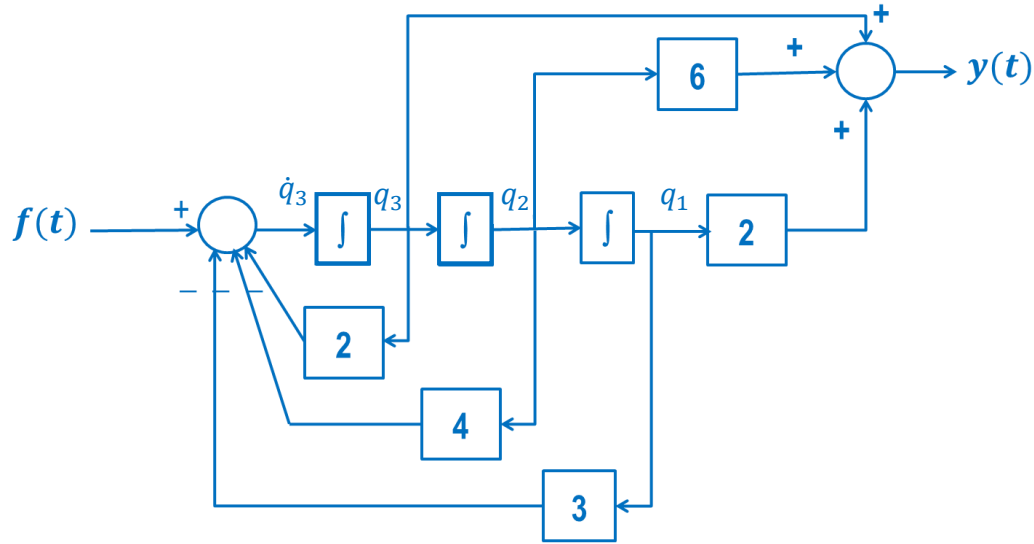
The system model has 3 state variables, 1 input, and 1 output.

Form the state equations and the output equations in the standard matrix-vector form.

$$\text{State Equation} \quad \rightarrow \quad \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -4 & -2 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} f(t)$$

$$\text{Output Equation} \quad \rightarrow \quad y(t) = \begin{bmatrix} 2 & 6 & 1 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} f(t)$$

The block diagram to visualize the state variables, input, and output:

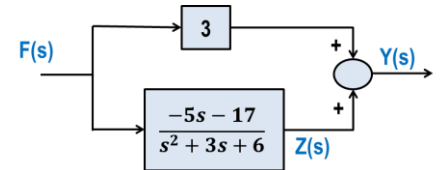


e) $\frac{Y(s)}{F(s)} = \frac{3s^2 + 4s + 1}{s^2 + 3s + 6}$

This is a **proper** transfer function.

First, we have to rewrite it as a summation of a **constant term** and a **strictly proper** function.

$$\frac{Y(s)}{F(s)} = \frac{3s^2 + 4s + 1}{s^2 + 3s + 6} = 3 + \frac{-5s - 17}{s^2 + 3s + 6}$$



The feed-forward matrix **D** is obtained as the constant term 3.

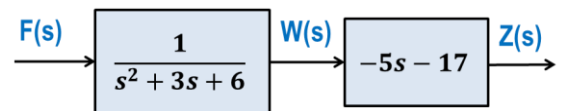
Then, determine the matrices **A**, **B**, and **C** from the strictly proper transfer function.

$$\frac{Z(s)}{F(s)} = \frac{-5s - 17}{s^2 + 3s + 6}$$

This is a **strictly proper** transfer function with a **polynomial** in the numerator.

Since, the numerator is a polynomial of s , we have to split it into two parts as below:

$$\frac{Z(s)}{F(s)} = \frac{-5s - 17}{s^2 + 3s + 6} = \left(\frac{1}{s^2 + 3s + 6} \right) (-5s - 17)$$



First, find the state equation from the part with the denominator:

$$\frac{W(s)}{F(s)} = \frac{1}{s^2 + 3s + 6}$$

$$s^2 W(s) + 3s W(s) + 6W(s) = F(s) \quad \rightarrow \quad \ddot{w}(t) + 3\dot{w}(t) + 6w(t) = f(t)$$

Define the state variables:

$$q(t) = w(t)$$

$$q_2(t) = \dot{w}(t)$$

Find the first derivative of the state variables and rewrite them in terms of the state variables and the input.

$$\dot{q}_1(t) = \dot{w}(t) \rightarrow \dot{q}_1(t) = q_2(t)$$

$$\dot{q}_2(t) = \ddot{w}(t) \rightarrow \dot{q}_2(t) = -3\dot{w}(t) - 6w(t) + f(t) = -3q_2(t) - 6q_1(t) + f(t)$$

Find the output equation by considering the effect of the block with the numerator.

$$Z(s) = (-5s - 17)W(s) \rightarrow Z(s) = -5sW(s) - 17W(s)$$

$$\rightarrow z(t) = -5\dot{w}(t) - 17w(t) = -5q_2(t) - 17q_1(t)$$

Therefore, the output equation is:

$$y(t) = z(t) + 3f(t) = -5q_2(t) - 17q_1(t) + 3f(t)$$

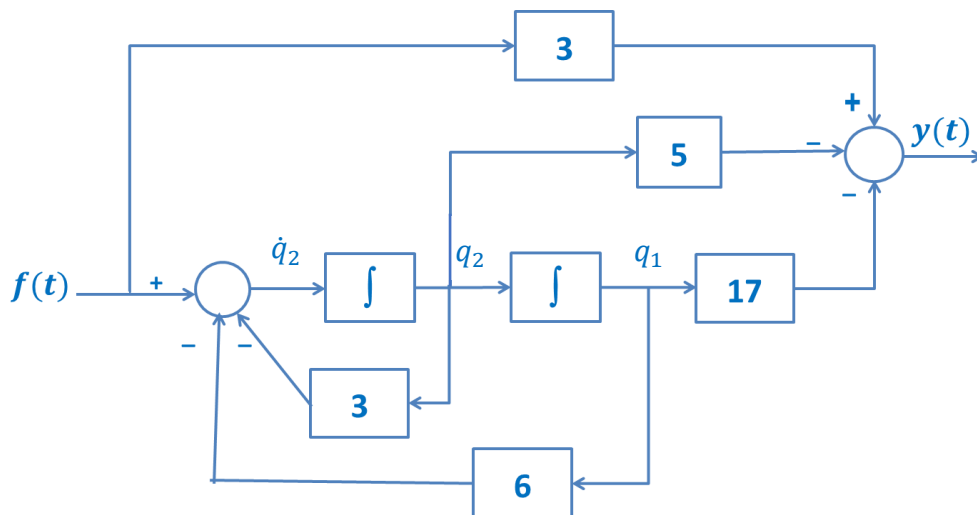
The system model has 2 state variables, 1 input, and 1 output.

Form the state equations and the output equations in the standard matrix-vector form.

$$\text{State Equation} \rightarrow \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -3 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f(t)$$

$$\text{Output Equation} \rightarrow y(t) = [-17 \quad -5] \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + [3]f(t)$$

The block diagram to visualize the state variables, input, and output:

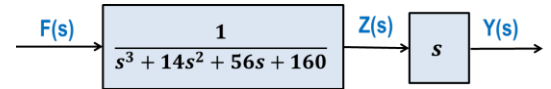


f) $\frac{Y(s)}{F(s)} = \frac{s}{s^3 + 14s^2 + 56s + 160}$

This is a **strictly proper** transfer function with a **polynomial** in the numerator.

Since, the numerator is a polynomial of s , we have to split it into two parts as below:

$$\frac{Y(s)}{F(s)} = \frac{s}{s^3 + 14s^2 + 56s + 160} = \left(\frac{1}{s^3 + 14s^2 + 56s + 160} \right) (s)$$



First, find the state equation from the part with the denominator:

$$\frac{Z(s)}{F(s)} = \frac{1}{s^3 + 14s^2 + 56s + 160}$$

$$s^3 Z(s) + 14s^2 Z(s) + 56s Z(s) + 160 Z(s) = F(s) \rightarrow \ddot{z}(t) + 14\dot{z}(t) + 56\dot{z}(t) + 160z(t) = f(t)$$

Define the state variables:

$$q_1(t) = z(t)$$

$$q_2(t) = \dot{z}(t)$$

$$q_3(t) = \ddot{z}(t)$$

Find the first derivative of the state variables and rewrite them in terms of the state variables and the input.

$$\dot{q}_1(t) = \dot{z}(t) \rightarrow \dot{q}_1(t) = q_2(t)$$

$$\dot{q}_2(t) = \ddot{z}(t) \rightarrow \dot{q}_2(t) = q_3(t)$$

$$\dot{q}_3(t) = \ddot{z}(t) \rightarrow \dot{q}_3(t) = -14\ddot{z}(t) - 56\dot{z}(t) - 160z(t) + f(t) = -14q_3(t) - 56q_2(t) - 160q_1(t) + f(t)$$

Find the output equation by considering the effect of the block with the numerator.

$$Y(s) = sZ(s) \rightarrow y(t) = \dot{z}(t) = q_2(t)$$

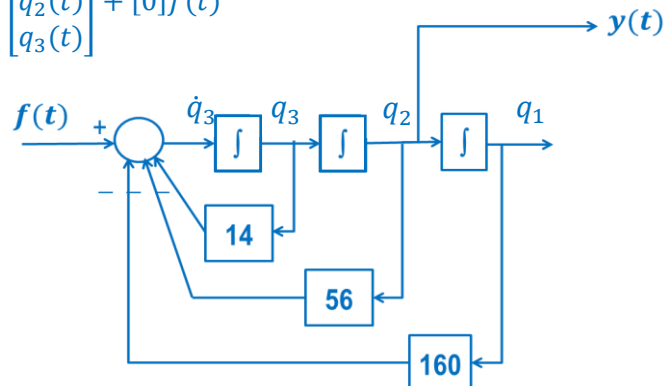
The system model has 3 state variables, 1 input, and 1 output.

Form the state equations and the output equations in the standard matrix-vector form.

$$\text{State Equation} \rightarrow \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -160 & -56 & -14 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} f(t)$$

$$\text{Output Equation} \rightarrow y(t) = [0 \quad 1 \quad 0] \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + [0] f(t)$$

The block diagram to visualize the state variables, input, and output:



g) $\frac{Y(s)}{F(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$

This is a **strictly proper** transfer function with a **constant value** in the numerator.

First, find the associate differential equation:

$$s^3 Y(s) + 9s^2 Y(s) + 26s Y(s) + 24Y(s) = 24F(s) \rightarrow \ddot{y}(t) + 9\dot{y}(t) + 26\dot{y}(t) + 24y(t) = 24f(t)$$

Define the state variables:

$$q_1(t) = y(t)$$

$$q_2(t) = \dot{y}(t)$$

$$q_3(t) = \ddot{y}(t)$$

Find the first derivative of the state variables and rewrite them in terms of the state variables and the input.

$$\dot{q}_1(t) = \dot{y}(t) \rightarrow \dot{q}_1(t) = q_2(t)$$

$$\dot{q}_2(t) = \ddot{y}(t) \rightarrow \dot{q}_2(t) = q_3(t)$$

$$\dot{q}_3(t) = \ddot{y}(t) \rightarrow \dot{q}_3(t) = -9\dot{y}(t) - 26\dot{y}(t) - 24y(t) + 24f(t) = -9q_3(t) - 26q_2(t) - 24q_1(t) + 24f(t)$$

Find the output in terms of the state variables and the input.

$$y(t) = q_1(t)$$

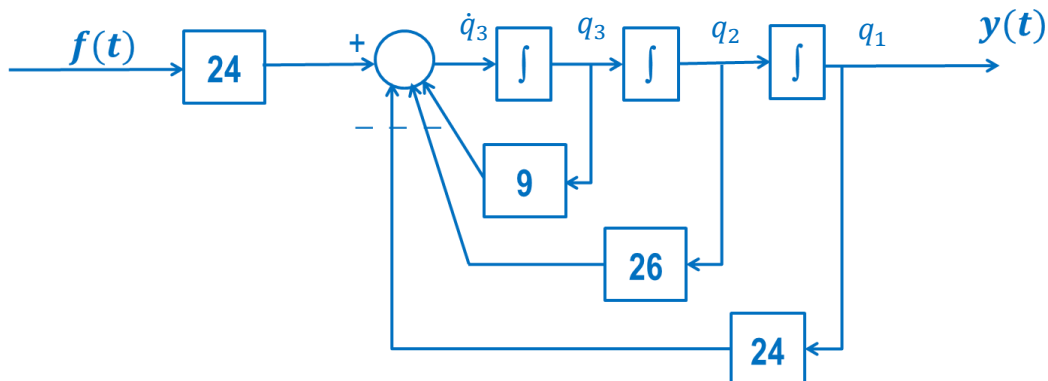
The system model has 3 state variables, 1 input, and 1 output.

Form the state equations and the output equations in the standard matrix-vector form.

$$\text{State Equation} \rightarrow \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} f(t)$$

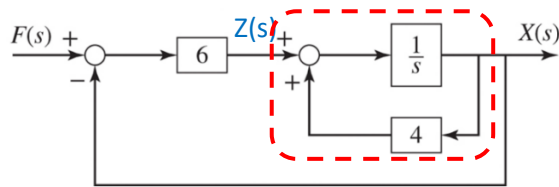
$$\text{Output Equation} \rightarrow y(t) = [1 \ 0 \ 0] \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + [0]f(t)$$

The block diagram to visualize the state variables, input, and output:



PART 5: Block Diagram Models with Transfer Function

1) Obtain the transfer function $X(s)/F(s)$ for each of the following block diagrams.



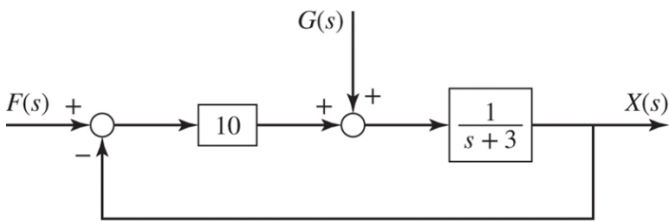
a)

First find the transfer function of internal feedback. Note that it is a positive feedback loop,

$$\frac{X(s)}{Z(s)} = \frac{\frac{1}{s}}{1 - \frac{4}{s}} = \frac{\frac{1}{s}}{\frac{s-4}{s}} = \frac{1}{s-4}$$

Then find the overall transfer function, which is a negative feedback loop,

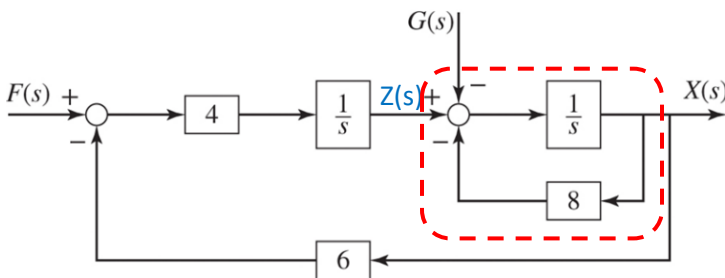
$$\frac{X(s)}{F(s)} = \frac{6 \left(\frac{1}{s-4} \right)}{1 + 6 \left(\frac{1}{s-4} \right)} = \frac{\frac{6}{s-4}}{\frac{s-4+6}{s-4}} = \frac{6}{s+2}$$



b)

This is a two-input one-output system. From the superposition principles, assume that $G(s) = 0$ and solve for $X(s)/F(s)$, which is a negative feedback loop,

$$\frac{X(s)}{F(s)} = \frac{10 \left(\frac{1}{s+3} \right)}{1 + 10 \left(\frac{1}{s+3} \right)} = \frac{\frac{10}{s+3}}{\frac{s+3+10}{s+3}} = \frac{10}{s+13}$$



c)

This is a two-input one-output system. From the superposition principles, assume that $G(s) = 0$ and solve for $X(s)/F(s)$:

First find the transfer function of internal feedback, which is a negative feedback loop:

$$\frac{X(s)}{Z(s)} = \frac{\frac{1}{s}}{1 + \frac{8}{s}} = \frac{\frac{1}{s}}{\frac{s+8}{s}} = \frac{1}{s+8}$$

Then find the overall transfer function,

$$\frac{X(s)}{F(s)} = \frac{4\left(\frac{1}{s}\right)\left(\frac{1}{s+8}\right)}{1 + 4\left(\frac{1}{s}\right)\left(\frac{1}{s+8}\right)(6)} = \frac{\frac{4}{s(s+8)}}{1 + \frac{24}{s(s+8)}} = \frac{\frac{4}{s(s+8)}}{\frac{s(s+8) + 24}{s(s+8)}} = \frac{4}{s^2 + 8s + 24}$$

2) Draw block diagram for the following equation. The output is $X(s)$, the inputs are $F(s)$ and $G(s)$.

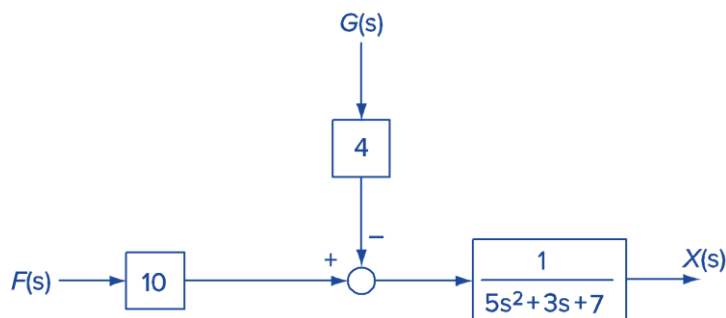
$$5\ddot{x}(t) + 3\dot{x}(t) + 7x(t) = 10f(t) - 4g(t)$$

Take Laplace transform:

$$5s^2X(s) + 3sX(s) + 7X(s) = 10F(s) - 4G(s)$$

Solve for $X(s)$:

$$X(s) = \frac{10}{5s^2 + 3s + 7}F(s) - \frac{4}{5s^2 + 3s + 7}G(s)$$



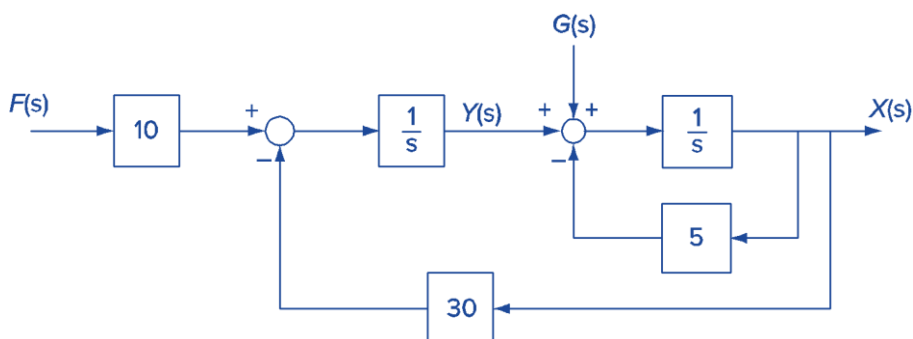
3) Draw the block diagram for the following model. The output is $X(s)$, the inputs are $F(s)$ and $G(s)$. Indicate the location of $Y(s)$ on the diagram.

$$\dot{x}(t) = y(t) - 5x(t) + g(t), \quad \dot{y}(t) = 10f(t) - 30x(t)$$

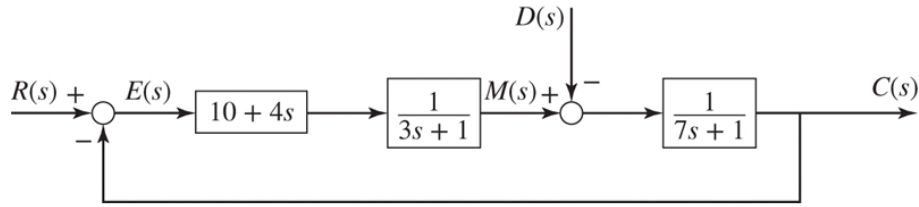
Take Laplace transform from each equation:

$$sX(s) = Y(s) - 5X(s) + G(s) \rightarrow X(s) = \frac{1}{s}[Y(s) - 5X(s) + G(s)]$$

$$sY(s) = 10F(s) - 30X(s) \rightarrow Y(s) = \frac{1}{s}[10F(s) - 30X(s)]$$



4) Given the following block diagram, derive the expression for the variables $C(s)$, $E(s)$, and $M(s)$ in terms of $R(s)$ and $D(s)$.



This is a two-input one-output system. Apply superposition principles to find the $C(s)$.

Assume $G(s) = 0$, solve for $C(s)/R(s)$

$$\frac{C(s)}{R(s)} = \frac{(10 + 4s) \left(\frac{1}{3s + 1} \right) \left(\frac{1}{7s + 1} \right)}{1 + (10 + 4s) \left(\frac{1}{3s + 1} \right) \left(\frac{1}{7s + 1} \right)} = \frac{\frac{4s + 10}{(3s + 1)(7s + 1)}}{\frac{(3s + 1)(7s + 1) + 10 + 4s}{(3s + 1)(7s + 1)}} = \frac{4s + 10}{21s^2 + 14s + 11}$$

Assume $R(s) = 0$, solve for $C(s)/D(s)$

$$\frac{C(s)}{D(s)} = \frac{-\left(\frac{1}{7s + 1} \right)}{1 + (10 + 4s) \left(\frac{1}{3s + 1} \right) \left(\frac{1}{7s + 1} \right)} = \frac{\frac{-1}{7s + 1}}{\frac{(3s + 1)(7s + 1) + 10 + 4s}{(3s + 1)(7s + 1)}} = \frac{-(3s + 1)}{21s^2 + 14s + 11}$$

Therefore, the $C(s)$ is obtained in terms of $R(s)$ and $D(s)$

$$C(s) = \frac{4s + 10}{21s^2 + 14s + 11} R(s) - \frac{3s + 1}{21s^2 + 14s + 11} D(s)$$

From the block diagram:

$$E(s) = R(s) - C(s) = R(s) - \frac{4s + 10}{21s^2 + 14s + 11} R(s) + \frac{3s + 1}{21s^2 + 14s + 11} D(s)$$

$$E(s) = \frac{21s^2 + 10s + 1}{21s^2 + 14s + 11} R(s) + \frac{3s + 1}{21s^2 + 14s + 11} D(s)$$

From the block diagram:

$$M(s) = \frac{4s + 10}{3s + 1} E(s) = \frac{(4s + 10)(7s + 1)}{21s^2 + 14s + 11} R(s) + \frac{4s + 10}{21s^2 + 14s + 11} D(s)$$